Cyberdisc rotating-collider admission

Created-at: 2026-10-05 03:21 CEST. ETA: 25 minutes (focused regression, small owner fix, targeted checks and report). No benchmark or simulation campaign.

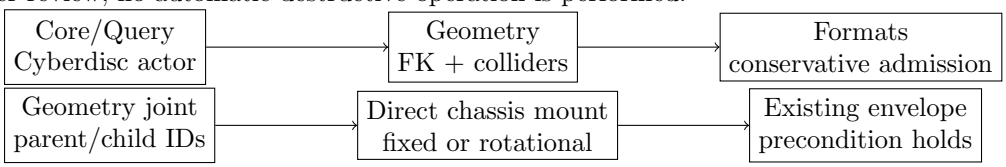
Owner and scope Transformation/Formats owns the generated-world actor admission. Geometry owns collision primitives and forward kinematics; Robotics owns the unchanged actor. Reuse these owners. A collider offset may point inward at zero wheel angle and outward after half a revolution. Replace the zero-pose centre norm with the conservative triangle bound: distance from chassis to link origin plus offset norm plus existing shape bound.

Acceptance First reproduce old admission of the shifted-wheel fixture, while existing FK shows its half-turn envelope exceeds the cell clearance. The fixed admission rejects it. The unchanged zero-offset actor retains the exact previous envelope and the existing small-world export passes. This bounds collider rotation about its link; no general articulated sweep or dynamic route-following guarantee is claimed.

Files Existing `hymeko_formats/examples/adversarial_grid/physical.rs`; this plan bundle; `reports/2026-10-05-rotatin` and its raw logs. CORE.YAML items touched: []. No public signatures, schema, dependency, actor, controller or physics changes. Existing 64-obstacle, 32-waypoint, 2m-cell caps and 0.16m arrival tolerance remain unchanged.

Checks and budget Pure FK regression on local Rust, exact ignored sibling actor integration tests, ordinary grid correctness tests and Clippy. No rollout, inference, performance measurement or generated-world ranking. At most two build jobs; each check bounded externally to 180s where practical. No package installation, worktree, commit or push. Worst-case accepted profile remains 16x16 cells, 64 obstacles and 32 waypoints.

Risks Conservative triangle inequality may refuse an offset actor that would fit; this is explicit admission conservatism. Bounds rely on the existing shape owner and reusable Cyberdisc layout, not arbitrary articulated chains. Existing zero offsets must preserve the old result exactly. Rollback consists of reverting this isolated source delta after review; no automatic destructive operation is performed.



Continuation: collider mount assumptions

Created-at: 2026-10-05 21:13 CEST. ETA: 25 minutes, based on two pure FK counterexamples, a private admission check, regression tests, and reporting.

Hypothesis The offset norm bound assumes that each collider link's origin remains at fixed distance from the chassis. The admitted source currently permits nesting a wheel under another driven wheel or making its mount prismatic. At zero configuration these may fit, then leave the corridor under joint motion.

Discriminating check Compile controlled changes to the existing actor; use owner FK and support queries at zero and one nonzero joint value. Preserve before-fix false admission in logs. If reproduced, require every non-chassis collision-bearing link to have a direct fixed/rotational mount on the chassis, using canonical declaration IDs and projected JointType. This checks the actual existing Cyberdisc profile; no general robot importer or articulated sweep is added.

Acceptance and boundaries Both invalid mounts fail before output creation. Existing actor/export, shifted-offset regression, and grid tests pass. Clippy, format, diff and provider duplication checks apply. CORE.YAML: []. Same source file, this existing plan/report, and a new mount-admission evidence subfolder. No dependency, schema, controller or engine changes. No Mission Control access. No simulation is needed to establish static geometric counterexamples; any later simulation remains Amaterasu-only. Two build jobs, 180-second command ceiling, no benchmark/profiling, no commit/push/merge/worktree. Preserve prior dirty edits.

Risk This intentionally refuses nested and translating collider mounts, including configurations that could fit. Fixed/rotational direct mounts are sufficient for the current envelope; arbitrary actors remain outside scope. Rollback removes only this continuation's reviewed delta, preserving the earlier fix.